Designing Specific Tasks with Structured Prompts - Practice

Lab Objective

Tomasterthe prompt engineering cyclebysystematically assemblingprompt elements and tuning the results for optimal quality

Lab Overview

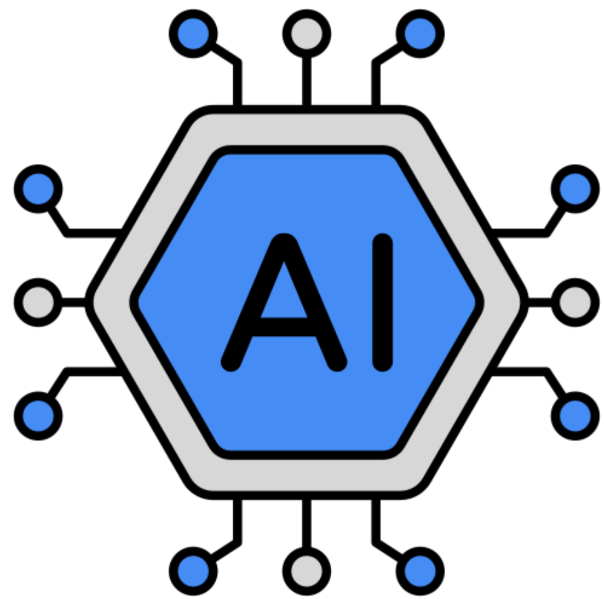
Lab 2-1: Step-by-Step Prompt Element Practice (10 min)

Lab 2-2: CreatingMy Own AI Study Coach with Prompt Engineering (10 min)

Lab 2-3: Structured Prompt Design (10 min)

Lab 2-4: ModelExecution and 1st Result Check (10 min)

Lab 2-5: Prompt1stTuning and Re-execution (20 min)



Lab 2-1: **Step-by-Step Prompt Element Practice (10 min)**

Mission

Build a Professional Health Coach to explore how totransforms a vague request into a high-quality, actionable AI response.

Discussion Point

After completing Step 6, thinkabout the following.

Points

1. In which Step did you feel the quality of the response changed the most? Why?
2. Which two elements do you think are the most "must-have" for your daily tasks?
3. If the AI gives a disappointing answer next time, which element will you tune first?

Activity

Pick one representative result from Step 0-6 and share your insights with class:

“I felt the biggest 'quality jump' at Step [4], because of the constraints i guess. When i give spesific constraints about my life balance to Gemini it gives more appropriate and smart advice.
Also, The element I will use most often is [CONSTRAINTS] because it makes the AI more efficient and provides outputs exactly according to my needs. “

Lab 2-2: Creating My Own AI Study Coach with PromptEngineering(10min)

Mission

Design an AI Study Coach that generates a realistic, adherence-focused study plan based on a scenario.

Discussion Point

To solve the student's problem of "hard to stick to plans,“
what is the most important Constraint or Instruction you should give to the AI?

Example

1. **Buffer Time:** How can we build "intentional flexibility" into the schedule to handle unexpected delays?
2. **Specific Breaks:** What is the optimal timing and style of rest to maximize long-term focus?
3. **Priority Logic:** What clear rules should the AI follow to re-rank subjects when your daily priorities shift?

Activity

Share your strategy with the class:

“I think adding a priority logic is key to making this plan realistic because in the real world the plan never executes perfectly. If we have another plan for tracking our tasks according to priority level at least we can complete the most important ones."

Lab 2-3: Structured Prompt Design (10 min)

Mission

Crafting the Master Template by organizing study coach instructions into a clear, hierarchical structure that the AI can parse with high accuracy.

Discussion Point

Reflect on the following questions to build the logic for your template. How does each section "control" the AI?

Points

1. Why did you choose this specific Role to solve the student's problem?
2. How does the Goal focus the AI's attention on "execution" rather than just "information"?
3. Which Constraint directly addresses the "hard to stick to plans" issue?
4. How does the Verification step force the AI to "double-check" its own logic?

Activity

Pick the most important sections of your template and explain your logic:

"I intentionally designed the Flexible & Personal Constraints to be flexible because on the grounds that there might also be students working part-time. The reason I chose "when u finished the plan control it, is it compatible with constraints or not" for the Verification was to prevent the AI from hallusination."

Lab 2-4: Model Execution and1stResultCheck(10min)

Mission

Test your structured prompt and evaluate the AI's response based on the "Reality Rules" you defined in the template.

Discussion Point

After generating the first result, review the output critically withfollowing.

Points

1. **In which Section** did you feel the **quality** of the response changed the most compared to a simple prompt? Why?
2. Which **two elements** from your [Constraints] or [Style] do you think were the most "must-have" for making the plan realistic?
3. If the AI gives a disappointing answer next time, which specific element ([Negative Prompt] or [Verification]) will you **tune first**?

Activity

Share your evaluation and "Tuning Plan" with the class:

Initially, the AI claimed it followed the priority rule but failed to provide a clear 'Plan B' for days when I am too tired to study,so I plan to refinethe[Verification] by forcing it to clearly write down a 'Task Emergent Priority List' at the end of the plan to improve accuracy.

Lab 2-5: Prompt 1st Tuning andRe-execution(20min)

Mission

Refine your template based on the evaluation from Lab 2-4 to achieve the highest level of adherence and realism in the study plan.

Discussion Point

Reflect on the "Gap" between your design intent and the AI's actual performance.

Points

1. **Which specific instruction did the AI struggle with the most, and what do you think was the cause of this "instruction drift"?**
2. If you were to add one more **[Negative Prompt]** to prevent a common mistake, what would it be?
3. How did the **[Verification]** step help (or fail) in catching errors during the first execution?

Activity

Share your final refinement strategy with the class:
"By adjusting the Verification & Constraints, I wasable to solvetheproblem of the AI not providing a backup plan for my tired days, resulting in a plan that is truly flexible, realistic, and burnout-proof.